Git & GitHub

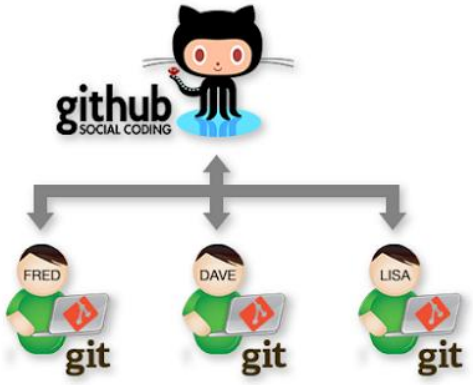
Git

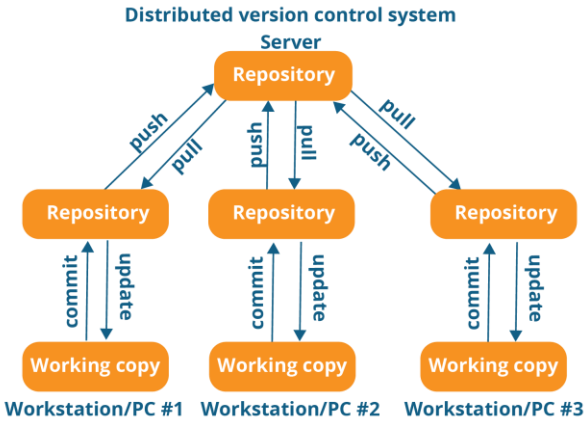
Git is a distributed version control system that helps developers track changes in code and collaborate on projects.

GitHub

GitHub is a cloud-based platform for hosting Git repositories, enabling collaboration and version control.

Git & GitHub Workflow





Steps for Git workflow:

1. **Initialize a repository.**
2. **Configure user information.**
3. **Stage files for a commit.**
4. **Commit changes locally.**
5. **Link to a remote repository.**
6. **Push changes to the remote repository.**
7. **Pull latest changed from remote repository.**
8. **Delete local repository (Optional)**

1. Create a New Local Git Repository

To initialize a new Git repository in a directory, navigate to the desired folder and execute the following command:

```
git init
```

2. Configure User Information (One-Time Setup)

Set your username and email for Git. This is typically a one-time setup for identifying the author of the commits. Replace "your name" and "your email" with your actual details:

```
git config --global user.name "your name"
git config --global user.email "your email"
```

3. Add Files or Folders to the Staging Area

To stage changes for the next commit, use the git add command. You can stage specific files, all files, or files matching certain patterns:

- **Add all files and folders to staging:**

```
git add -A
```

- **Add a specific file:**

```
git add filename
```

- **Add all files with a specific extension (e.g., Java files):**

```
git add *.py
```

- **Add all files within a folder:**

```
git add foldername
git add .
```

4. Commit Changes to the Local Repository

After staging the files, commit them to your local repository with a descriptive commit message:

```
git commit -m "commit message"
```

5. Connect Local Repository to a Remote Repository (One-Time Setup)

Link your local repository to a remote repository using the git remote add command. Replace the example URL with the actual remote repository URL:

```
git remote add origin " https://github.com/pavanoltraining/myproject.git"
```

6. Push Changes to the Remote Repository

To upload your committed changes to the remote repository, use the git push command. Specify the remote name (origin) and the branch name (master for the main branch):

```
git push origin master
```

Note: Need to pass token which is generated in GitHub.

7. Pull the latest Changes from the Remote Repository

To get latest changes from the remote repository to local repository , you need to use git pull command.

```
git pull origin master
```

8. To Delete local repository

```
rm -rf
```

Git Advanced Tutorials:

https://tinyurl.com/2ktjbscr